

Exploring House Prices in Ames, Iowa

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1. Introduction

For this project, I analyzed the Ames Housing dataset, which contains 2,930 residential property sales with features describing house size, quality, and SalePrice.

The goal of this analysis was not only to explore the data, but to understand the key drivers behind house pricing. Specifically, I aimed to answer three main questions:

- (1) What factors most influence house prices?
- (2) How do size and overall quality impact SalePrice?
- (3) Which features are most strongly correlated with the target variable?

Understanding these relationships is important because house prices are influenced by multiple interacting factors, and identifying the most important ones can help explain price variation and support future prediction models.

2. Cleaning Summary

The raw dataset contained several issues that needed to be addressed before analysis.

- **Missing values:** The Lot Frontage column had missing values, which were filled using the median. This approach was chosen because the presence of outliers makes the median more robust than the mean, ensuring that extreme values do not distort the imputation.
- **Categorical data:** Some categorical columns contained extra spaces and inconsistent formatting. These were cleaned to ensure that categories are treated correctly during analysis and encoding, avoiding duplicated or misleading categories.
- **Duplicates:** I checked for duplicate rows and found none, confirming that each observation represents a unique house sale.
- **Outliers:** A boxplot of SalePrice revealed extreme values above 400,000. Instead of removing them, I capped these values at the 99th percentile. This decision preserves the dataset size while reducing the influence of unusually high prices that could distort analysis results.

Finally, I validated the cleaned dataset by ensuring there were no missing values in key columns and that all SalePrice values were positive and valid. This step ensures the reliability of all subsequent analysis.

3. Feature Engineering Summary

To better capture the relationships in the data, I created several new features that provide more meaningful representations of house characteristics.

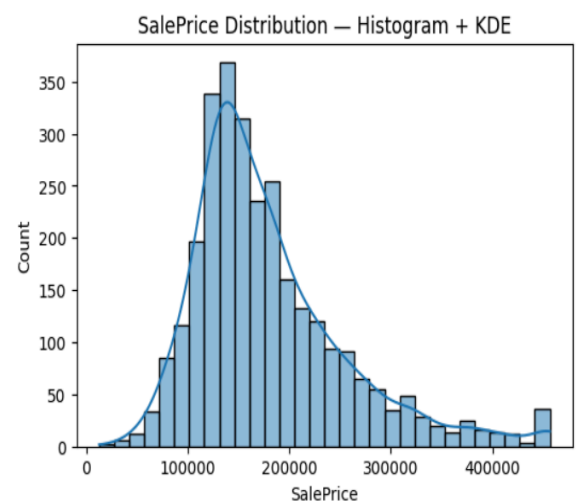
- **price_per_sqft:** This feature normalizes SalePrice by living area, allowing fair comparison between houses of different sizes. It helps identify whether a house is relatively expensive or cheap for its size, rather than relying on total price alone.
- **price_per_bedroom:** This feature provides insight into how price relates to the number of bedrooms. It helps reveal whether adding more bedrooms increases value proportionally or if some bedrooms add less economic value than expected.
- **quality_area:** This interaction feature (Overall Quality × living area) captures the combined effect of size and quality. It reflects the idea that a large, high-quality house is significantly more valuable than either size or quality alone, highlighting non-linear relationships in pricing.
- **Log transformation:** I applied a log transformation to SalePrice to reduce right skewness and stabilize variance. This makes the distribution more symmetric, which improves the reliability of analysis and helps models better capture relationships in the data.

Overall, these engineered features allow a deeper understanding of pricing patterns by capturing relative value, interaction effects, and distribution improvements that are not visible in the raw features alone.

4. Key Findings

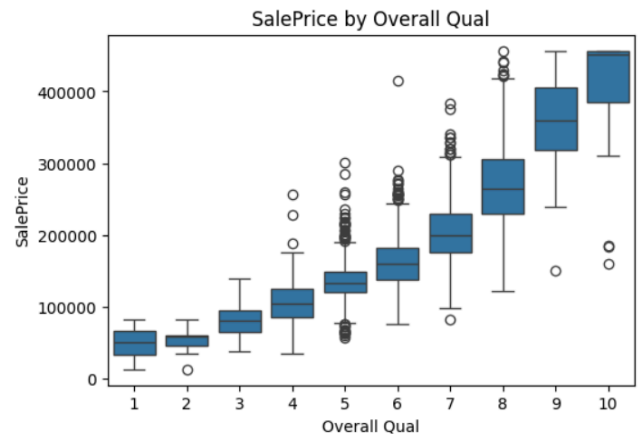
Finding 1 — Overall Quality strongly drives price differences.

The boxplot shows a clear increase in SalePrice as Overall Quality increases, with houses rated 8–10 having significantly higher median prices compared to those below 6. This suggests that construction quality, materials, and finishing level play a major role in pricing. The sharp jump between mid-quality and high-quality homes indicates that buyers value quality improvements disproportionately, meaning small increases in quality can lead to large increases in price.



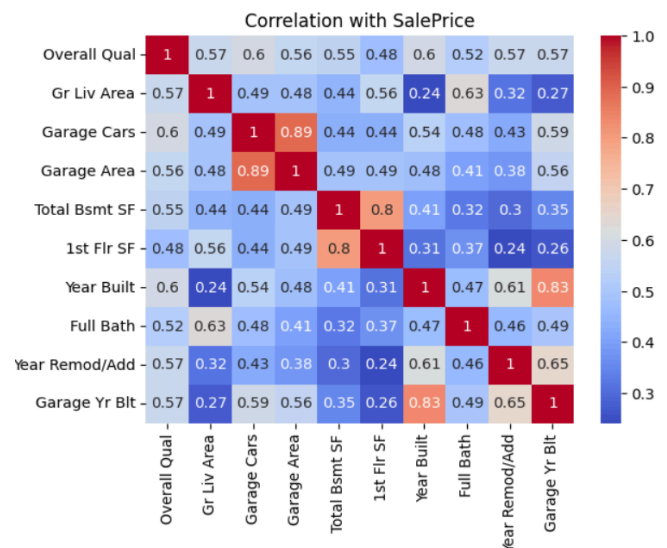
Finding 2 — Living area increases price, but not linearly.

The scatter plot shows that SalePrice increases as Gr Liv Area increases, especially between 1000 and 2500 sqft. However, beyond this range, the relationship becomes less consistent, and price increases slow down. This suggests diminishing returns, where adding more space does not always lead to proportional price growth. This could reflect market limits, where extremely large houses are less common or less demanded.



Finding 3 — Size and quality dominate price variation compared to other features.

The heatmap shows that Overall Quality and Gr Liv Area have the strongest correlations with SalePrice, while other variables contribute less significantly. This indicates that most of the variation in house prices can be explained by these two factors. The weaker correlations of other features suggest they play a secondary role, meaning a model focusing on size and quality would already capture a large portion of price behavior.



7. What I Would Do Next

If I had more time, I would build a Linear Regression model to predict SalePrice using the engineered features. I would evaluate the model using appropriate metrics such as RMSE to measure prediction accuracy.

In addition, I would explore more advanced feature engineering techniques and investigate interactions between variables to improve the model's performance. This would help better capture the complexity of factors influencing house prices.

This project was completed with the assistance of AI tools (ChatGPT) for explanation and debugging.